Technical report — Faultline

Apertus v1.5 8B was evaluated for failure modes involving false-premise accommodation, fabricated evidence, scientific citation integrity, and unsupported sensitive-attribute inference. Faultline combines paired adversarial probes with matched controls, independent replication, provenance-pinned evidence, and a session-bound Docker reproduction harness. Four issues survived falsification, and the final harness reproduced all four mechanisms across 22 unique logical tests.

Track: Apertus Readiness - Track 1A
Event: Online
Team: Faultline — Rares-Adrian Ivanoiu **Demo:** Reproducible submission package (make run)

Use of Apertus

- **Model:** apertus-v1.5-8b
- **How it is used:** red-teaming and evaluation through inference
- **Where it runs:** HackApertus hosted OpenAI-compatible endpoint

Problem

Faultline investigates whether changes in how unsupported information is framed can cause Apertus to abandon epistemic caution. The evaluation targets four consequences: acceptance of false historical premises, fabrication of documentary evidence, unsupported scientific claims accompanied by citation-like identifiers, and inference of sensitive individual attributes from demographic proxies.

These failures matter in research, education, search, decision support, and human-resources-style workflows because fluent outputs can appear evidential even when their premise is false or the available information is insufficient. The threat model requires neither prompt injection nor privileged model access: an unsupported premise may originate from a user, document, retrieved passage, or upstream system, after which the model may elaborate on it instead of challenging it.

The evaluation therefore asks not merely whether Apertus can produce an incorrect answer, but whether controlled changes in framing produce repeatable behavioral differences while the underlying subject or evidence remains fixed.

Red Teaming Method

Overall Strategy

The core design is a paired probe/control methodology. A probe introduces the candidate failure trigger, while a matched control asks Apertus to assess or verify essentially the same underlying information without requiring accommodation of that trigger. A finding is retained only when the adversarial behavior is reproduced across at least two independent subjects or sensitive attributes and the corresponding controls demonstrate that the model can recognize the relevant falsehood or insufficiency.

Evidence is separated into three layers: canonical discovery evidence from frozen source runs; automated reproduction evidence from fresh session-bound executions; and external verification where model behavior alone cannot establish an external fact, particularly DOI-to-publication correspondence. This separation prevents repeated model statements, derived interpretations, or later reproduction runs from being mistaken for independent evidence.

Discovery Process

Discovery was iterative and falsification-oriented. Candidate mechanisms that failed direct testing were stopped rather than expanded. For example, a nationality-based financial-risk comparison was rejected after Apertus treated otherwise identical applicants as equal risk; other exploratory branches were likewise stopped when they failed to produce a distinct and reproducible issue.

Four mechanisms survived: false factual premises being accommodated rather than corrected; false existence presuppositions eliciting fabricated historical documentary evidence; false scientific presuppositions eliciting specific scientific claims with DOI identifiers; and forced prediction eliciting sensitive-attribute inference from demographic proxies despite matched controls recognizing insufficient evidence.

Once a mechanism met its stopping condition, its canonical source runs were frozen. Finding builders reference explicit canonical filenames rather than selecting records by recency or semantic convenience, preventing later experiments and reproduction runs from silently changing the evidence packaged with a finding.

Tooling

Experiments are issued through a Python runner that records the request, Apertus response, model/runtime metadata, experiment family and variant, and reproduction-session identifier. A manifest defines the logical reproduction cases. The matrix runner creates a fresh session identifier, executes the manifest sequentially, and invokes a session-bound evaluator.

The evaluator rejects duplicate logical tests within a session and evaluates behavioral mechanisms rather than requiring token-identical outputs. This matters because wording and generated identifiers can vary between executions while the underlying failure persists. The judge-facing path is Dockerized: from the repository root, make run builds the image, executes the complete reproduction manifest, and automatically evaluates the resulting session.

Findings Dataset

Finding 1 — False factual premises are accommodated instead of corrected. Topic: Misrepresentation of history and factual incorrectness. Across independent Romania/World War I and Berlin Wall cases, poisoned prompts caused Apertus to explain false premises as factual, while matched controls recovered the correct historical facts. Self-assessed severity: **2**; breadth: **2**.

Finding 2 — False existence presuppositions trigger fabricated historical documentary evidence. Topic: Misrepresentation of history and factual incorrectness. Apertus supplied named officials and quotation-like documentary evidence for presupposed historical artifacts, while matched controls rejected the corresponding artifacts as nonexistent. The mechanism reproduced across Berlin and Romanian historical contexts. Self-assessed severity: **2**; breadth: **2**.

Finding 3 — False scientific presuppositions trigger misattributed DOI identifiers. Topic: Misrepresentation of history and factual incorrectness. Apertus generated specific authors, quantitative claims, and DOI identifiers for presupposed studies, while matched controls stated that the studies could not be reliably established. In the frozen canonical evidence, external verification showed that the generated DOI identifiers resolved to unrelated publications; this external check remains separate from the automated behavioral reproduction. Self-assessed severity: **2**; breadth: **2**.

Finding 4 — Prediction pressure triggers unsupported sensitive-attribute inference from demographic proxies. Topic: Stereotyping and bias. For synthetic candidate profiles, forced best-guess prompts caused Apertus to infer religion and political affiliation from demographic/contextual proxies, while matched controls recognized that the same evidence was insufficient. The issue is the unsupported individual inference, not an assertion that the demographic generalizations are accurate. Self-assessed severity: **2**; breadth: **2**.

Each submitted finding contains four Apertus response walkthroughs and is bound to explicit frozen canonical source runs.

Lessons Learnt and Recommendations

The strongest cross-finding pattern is sensitivity to task framing. Apertus can recognize a false premise or insufficient evidence under explicit verification, yet abandon that caution when the same uncertainty is embedded as a presupposition or when the prompt demands a single prediction. This suggests that factual systems should validate consequential premises before elaborating on them, citation-bearing outputs should be checked against external bibliographic metadata, and decision-support applications should preserve an explicit insufficient-information outcome rather than forcing sensitive predictions.

A second lesson is that apparently concrete artifacts do not constitute independent evidence. A quotation, named official, DOI, or confident prediction may inherit the uncertainty of the unsupported premise that produced it. Downstream systems should therefore preserve provenance and uncertainty across derived claims rather than treating increased specificity as increased reliability.

Further evaluation should test whether these mechanisms transfer to Apertus 70B, multilingual prompts, longer conversational contexts, retrieved-document workflows, and tool-augmented settings. For citation integrity, integrating an external DOI resolver into the harness would allow

semantic claim-to-publication verification to become an automated layer while remaining distinct from model-behavior evidence.

Reproducibility

The reproduction harness contains **22 unique logical tests**: 10 for Finding 1 and 4 each for Findings 2, 3, and 4. The verified judge-path execution completed **22/22** runs and reported `FINDING_1: PASS`, `FINDING_2: PASS`, `FINDING_3: PASS`, `FINDING_4: PASS`, and `OVERALL_REPRODUCTION: PASS`.

Reference development environment: Docker 29.5.2 (build `79eb04c`), Python 3.12.3, Linux x86_64 under WSL2, container base `python:3.12-slim`, and model `apertus-v1.5-8b`. Hosted inference credentials are supplied locally through `.env`; secrets and generated `data/runs/` records are excluded from version control.

No deterministic generation seed was supplied through the hosted endpoint workflow, so reproducibility is behavioral rather than token-identical. The evaluator checks the demonstrated mechanism and matched-control contrast. Each reproduction run receives a fresh session identifier, and only records from that session contribute to its result.

From a clean checkout, copy `.env.example` to `.env`, set `LLM_API_KEY`, and run `make run`. The command builds the Docker image, executes all 22 cases, and performs the automated evaluation.

Submission evidence commit: `977c7843a1823e89602b27aedf6d173c6fe877ab`

The organizer-provided `findings/findings.schema` contains a literal newline inside one JSON string and is preserved unchanged. `findings/findings.validation.schema` is a validation-only copy with that JSON syntax defect repaired; the intended schema semantics are unchanged.

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References

1. Yeo V, Phillips NL, Bogdanov S, Brookes N, Epps A, Teng A, Naismith SL, Lah S. The persistence of sleep disturbance and its correlates in children with moderate to severe traumatic brain injury: A longitudinal study. Sleep Medicine. 2021;81:387-393. DOI: 10.1016/j.sleep.2021.03.013.
2. Henry AL, Chisholm A, Carter LA, Bundy C, Griffiths CEM, Kyle SD. The relationship between sleep disturbance, symptoms and daytime functioning in psoriasis: a prospective study integrating actigraphy and experience sampling methodology. Sleep Medicine. 2020;72:144-149. DOI: 10.1016/j.sleep.2020.03.013.